

Solution Requirements

Date	15 July 2026
Team ID	*****
Project Name	SmartLender – Applicant Credibility Prediction for Loan Approval
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	User authentication is not required in the current version. Future versions can include login for bank officers or administrators.	Low
2	Authorization	Any applicant can submit loan details and receive a prediction. Future admin users can manage applications and models.	Low
3	External interfaces	The system accepts applicant information through a web form and predicts loan approval using a locally stored Random Forest model. No external APIs are used.	Medium
4	Transactions	Each submission follows a single workflow: user enters details → data preprocessing → model prediction → result display.	High
5	Reporting	The application displays the submitted details along with the predicted loan approval status. Future versions may include analytics dashboards.	Medium
6	Business rules	Applicant details are preprocessed exactly as during model training (encoding categorical values, scaling numerical values) before prediction.	High

7	Compliance to laws or regulations	Applicant data should be handled securely and used only for prediction purposes. Sensitive information should not be stored without user consent.	Medium
8	Other	The prediction should be generated quickly to provide a smooth user experience.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should provide loan predictions quickly.	Prediction generated in less than 1 second .
2	Scalability	The application should support multiple users simultaneously without significant performance degradation.	Supports 50+ concurrent users .
3	Security & Data Privacy	Applicant information should remain confidential and not be stored unnecessarily.	No sensitive applicant data stored permanently.
4	Reliability & Availability	The application should operate consistently without errors during demonstrations and normal usage.	Successfully completes the prediction workflow without crashes.
5	Usability & Accessibility	The user interface should be simple, responsive, and easy to navigate with properly labeled input fields.	All input fields clearly labeled and keyboard accessible.
6	Maintainability	The trained model, preprocessing steps, and encoding logic should be easy to update when retraining the model.	Model file and preprocessing code can be replaced with minimal code changes.